Internship Project

Title:
AI Driven Virtual Senior Mentor for Company Specific Preparation

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Introduction

Technical background of project

- The project is based on concepts from Machine Learning, Natural Language Processing (NLP), and Generative AI
- Uses text embeddings to convert resume and job descriptions into numerical vectors
- Applies similarity computation techniques to match resumes with relevant job roles
- Integrates Large Language Models (LLMs) to generate intelligent, human-like responses
- Built using Python-based ecosystem with libraries for data processing, vector computation, and AI interaction

Introduction

Technical Concepts (Algorithms) used

- Sentence Embeddings (via Transformer models)

→ Converts textual data into meaningful vector representations

- Cosine Similarity Algorithm

→ Measures similarity between resume and job description vectors

- Basic NLP Techniques

→ Text extraction, preprocessing, and keyword/skill identification

- Skill Gap Analysis Logic

→ Compares required skills vs available skills

- Prompt Engineering (LLM Context Design)

→ Provides structured input to generate relevant AI responses

Introduction

Motivation

- Students often lack clarity on:
- Suitable job roles based on their profile
- Required skills for specific companies
- Real interview processes
- Existing platforms:
- Provide generic guidance
- Do not personalize recommendations
- Motivation was to build a smart assistant that acts like a senior mentor and provides personalized, data-driven guidance

Introduction

Problem Statement

There is no unified system that:

- Analyzes resumes
- Matches them with job roles
- Identifies skill gaps
- Provides interview preparation guidance

Students rely on scattered sources, leading to:

- Inefficient preparation
- Lack of targeted improvement

The problem is to design a centralized intelligent system that bridges this gap.

Introduction

Area of application

- Campus Placement Preparation
- Career Guidance Platforms
- Resume Screening Systems
- EdTech Platforms

Dataset and Input Format

Resume Input:

- Format: PDF
- Extracted using text parsing tools

Job Descriptions Dataset:

- Stored as .txt files
- Contains role-specific requirements

Interview Experiences Dataset:

- Company-wise .txt files
- Includes real interview rounds and questions

User Query Input:

- Natural language text entered in chat interface

Literature Review

Cite Related work

Automated Resume Screening Systems

- Use keyword matching and basic NLP techniques
- Limitation: Lack of semantic understanding and personalization

Job Recommendation Systems

- Based on collaborative filtering and content-based filtering
- Focus on matching user profiles with job listings
- Limitation: Do not analyze skill gaps or provide improvement strategies

AI-based Career Guidance Platforms

- Provide general career advice using predefined rules or ML models
- Limitation: Not tailored to individual resumes and real interview data

Interview Preparation Platforms (e.g., LeetCode, Glassdoor)

- Provide coding problems and interview experiences
- Limitation: Lack integration with user profile and job matching

Recent Use of Transformer Models in NLP

- Models like BERT and Sentence Transformers used for semantic similarity
- Enable better understanding of textual meaning beyond keywords

Literature Review

Inference from Literature

Existing systems work in isolation:

- Resume screening
- Job recommendation
- Interview preparation

Lack of integration between:

- Resume analysis
- Skill gap detection
- Real interview insights

Traditional systems rely heavily on:

- Keyword matching
- Static rules

Modern approaches using transformer-based embeddings provide:

- Better semantic understanding
- Improved matching accuracy

Therefore, there is a need for a unified intelligent system that combines all these aspects.

Literature Review

SWOT analysis

Strengths

- Combines multiple functionalities in one system
- Uses semantic similarity (more accurate than keyword matching)
- Provides personalized AI-based guidance
- Incorporates real interview experiences

Weaknesses

- Depends on quality of input resume text
- Interview dataset is static (not real-time)
- Skill extraction is relatively basic
- Requires API usage (cost involved)

Opportunities

- Integration with real-time job portals
- Use of advanced NLP for better skill extraction
- Deployment as a web application for wider users
- Expansion to multiple domains (non-tech roles)

Threats

- Rapid advancements in AI tools
- Dependency on third-party APIs
- Data privacy concerns
- Competition from large-scale platforms

Objective

Main Objective

- To develop an AI-powered Virtual Senior Mentor that provides personalized placement guidance
- To integrate resume analysis, job matching, skill gap identification, and interview preparation into a single system
- To assist students in making data-driven career decisions and improving their chances of placement

Sub Objective

- To extract and process resume data from PDF format into structured text
- To implement semantic similarity using embeddings for accurate resume–job matching
- To identify missing skills by comparing resume content with job descriptions
- To design a system for storing and retrieving company-wise interview experiences
- To develop a mechanism for detecting company names from user queries
- To integrate a Large Language Model (LLM) for generating intelligent, personalized responses
- To build an interactive user interface using Streamlit for real-time interaction
- To ensure the system provides relevant, contextual, and actionable guidance

Methodology

Reference Software model

- The project follows an Incremental Development Model
- System is built in modular phases, where each module is developed and tested independently
- Key modules:
- Resume Processing
- Job Matching
- Skill Gap Analysis
- Interview Data Integration
- AI Response Generation

Advantages of this model:

- Easier debugging and testing
- Flexibility to improve individual components
- Faster development with continuous integration

Methodology

Steps

- The system follows a sequential pipeline:

Resume Upload

- User uploads resume in PDF format

Text Extraction

- Resume content is extracted using PDF parsing

Job Description Matching

- Resume is compared with multiple job descriptions using embeddings
- Best match and top matches are identified

Skill Gap Analysis

- Missing skills are identified by comparing resume and job requirements

Interview Data Retrieval

- Company-specific interview experiences are fetched

User Query Processing

- User asks placement-related questions

AI Response Generation

- Context (resume + skills + interview data) is passed to LLM
- Personalized response is generated

Methodology

Timeline

- Phase 1: Problem Understanding & Requirement Analysis
- Phase 2: Resume Processing & Data Preparation
- Phase 3: Implementation of Resume–JD Matching
- Phase 4: Skill Gap Analysis Module Development
- Phase 5: Interview Data Integration
- Phase 6: LLM Integration for AI Responses
- Phase 7: UI Development using Streamlit
- Phase 8: Testing & Debugging

Methodology

Deliverable of each steps or phase

- Phase 1: Defined problem statement and system design
- Phase 2: Working resume text extraction module
- Phase 3: Functional job matching system with similarity scores
- Phase 4: Skill gap detection output
- Phase 5: Structured interview experience dataset
- Phase 6: AI-based mentor response system
- Phase 7: Interactive user interface
- Phase 8: Fully integrated and tested application

Working Model

Requirement analysis (Link of SRS)

- The system is designed based on the following key requirements:

Functional Requirements

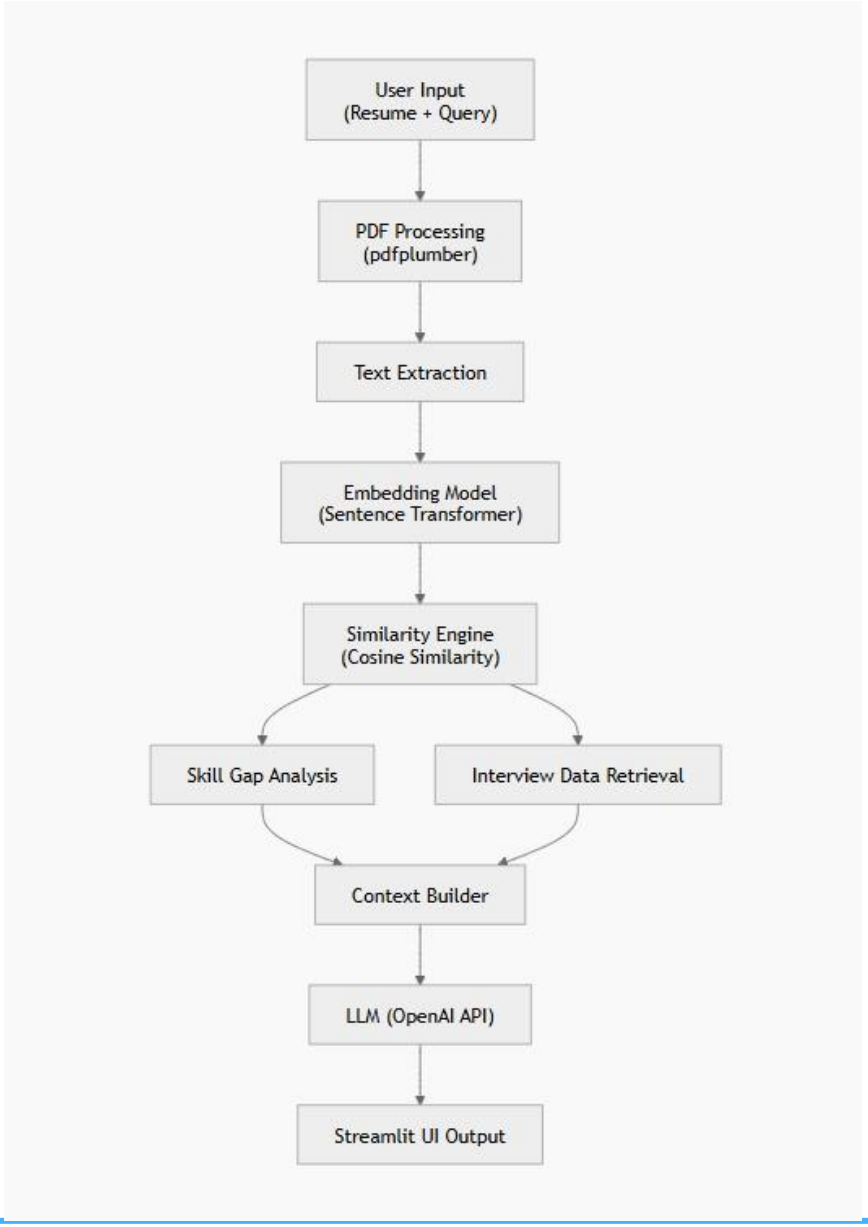
- Upload resume in PDF format
- Extract and process resume text
- Match resume with multiple job descriptions
- Identify missing skills
- Retrieve company-specific interview experiences
- Generate AI-based responses for user queries

Non-Functional Requirements

- Fast response time
- User-friendly interface
- Scalable architecture
- Secure handling of API keys and data

Working Model

Technical Diagram



Working Model

Working Module

- Step-by-step working:
- Resume Upload
- User uploads resume via UI
- Text Extraction
- Resume PDF is converted into text
- Embedding Generation
- Resume and job descriptions are converted into vectors
- Similarity Calculation
- Cosine similarity is used to find best matching job roles
- Skill Gap Analysis
- Missing skills are identified by comparing resume and JD
- Company Detection
- User query is analyzed to detect target company
- Interview Data Fetching
- Relevant interview experiences are retrieved
- Context Preparation
- Resume + skills + interview data combined
- AI Response Generation
- LLM generates personalized answer

Working Model

Attained Deliverables

- Fully functional AI-based placement mentor system
- Accurate resume–job matching module
- Effective skill gap identification system
- Integrated interview experience database
- Working AI chatbot for placement guidance
- Interactive user interface using Streamlit

Results

Tests Cases

Test Case 1: Resume Upload & Processing

- Input: PDF resume
- Expected Output: Text extracted successfully
- Result: Successfully extracted multi-page resume content

Test Case 2: Resume–JD Matching

- Input: Resume text + Job Descriptions
- Expected Output: Best matching job role with similarity score
- Result: System correctly identified top matching companies with similarity percentages

Test Case 3: Skill Gap Analysis

- Input: Resume + Best Matched JD
- Expected Output: List of missing skills
- Result: System identified missing skills such as Data Structures, Algorithms, C++

Test Case 4: Interview Data Retrieval

- Input: User query (e.g., “Prepare for Amazon”)
- Expected Output: Company-specific interview experiences
- Result: System successfully retrieved relevant interview rounds and questions

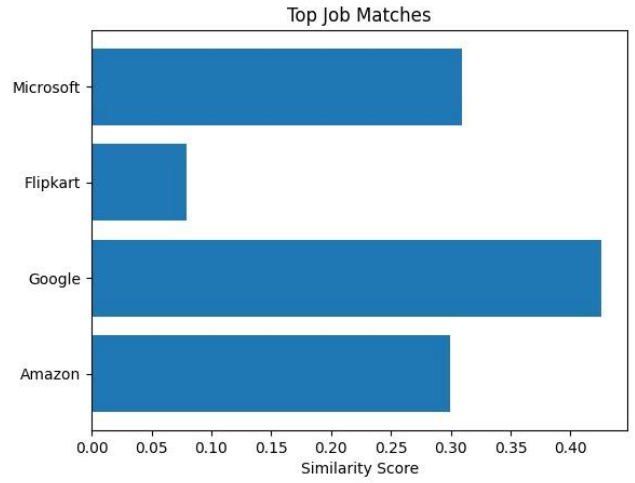
Test Case 5: AI Response Generation

- Input: User query + context
- Expected Output: Personalized guidance
- Result: LLM generated structured and relevant preparation strategies

Results

Outcome Graphs

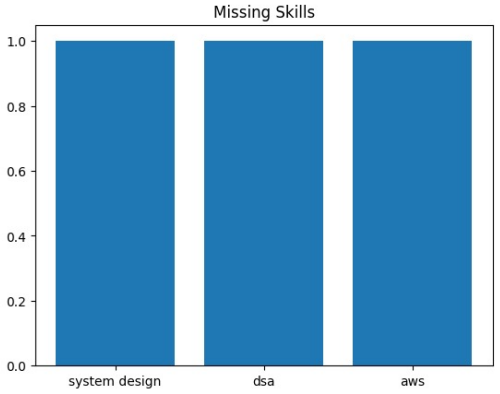
```
plt.figure()
plt.barh(labels, scores)
plt.title("Top Job Matches")
plt.xlabel("Similarity Score")
plt.show()
```




```
resume_skills = set(["python", "machine learning", "sql", "data structures"])
jd_skills = set(["python", "dsa", "system design", "aws"])

missing_skills = list(jd_skills - resume_skills)

plt.figure()
plt.bar(missing_skills, [1]*len(missing_skills))
plt.title("Missing Skills")
plt.show()
```



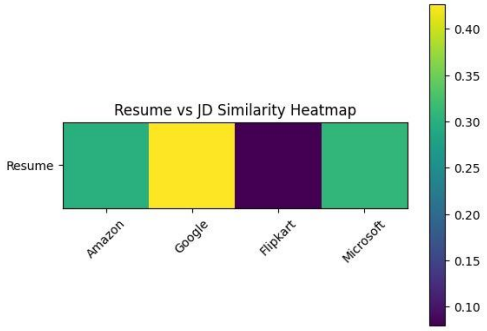

```
heatmap_data = np.array(scores).reshape(1, -1)

plt.figure()
plt.imshow(heatmap_data)
plt.colorbar()

plt.xticks(range(len(labels)), labels, rotation=45)
plt.yticks([0], ["Resume"])

plt.title("Resume vs JD Similarity Heatmap")

plt.show()
```



Results

Comparative Studies

Comparison with Traditional Methods:

Feature	Traditional Systems	Proposed System
Resume Matching	Keyword-based	Semantic (Embeddings)
Skill Analysis	Limited	Detailed Gap Detection
Interview Prep	Separate platforms	Integrated
Personalization	Low	High
AI Guidance	Not available	Available

Conclusion

Justification of Objectives

The primary objective of developing an AI Virtual Senior Mentor has been successfully achieved

The system effectively integrates:

- Resume analysis
- Job matching using semantic similarity
- Skill gap identification
- Interview experience retrieval
- AI-based personalized guidance

The use of embedding techniques improved the accuracy of job matching compared to traditional keyword-based methods

The system provides context-aware and personalized responses using LLM integration

All functional modules were successfully implemented and tested, ensuring end-to-end working of the system

Overall, the project successfully addresses the problem of fragmented placement preparation by providing a unified intelligent solution

Conclusion

Future Scope

- Integration with real-time job portals (e.g., LinkedIn, Naukri)
- Implementation of advanced NLP techniques for better skill extraction and analysis
- Expansion of interview dataset with real-time updates and larger coverage
- Development of a full-scale web application with backend APIs (Flask/FastAPI)
- Addition of user profile tracking and progress monitoring system
- Support for multiple domains beyond technical roles
- Integration of voice-based interaction and chatbot enhancements

Reference

[1] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” EMNLP, 2019.

🔗 View Paper (SBERT)

➞ Used for semantic similarity between student profiles and job descriptions.

[2] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL-HLT, 2019.

🔗 View Paper (BERT)

[3] D. Lavi, V. Medentsiy, and D. Graus, “conSultantBERT: Matching Jobs and Job Seekers using Sentence-BERT,” arXiv, 2021.

🔗 https://arxiv.org/abs/2109.06501

➞ Directly related to job-resume matching system.

[4] T. Mikolov et al., “Efficient Estimation of Word Representations in Vector Space,” arXiv, 2013.

🔗 https://arxiv.org/abs/1301.3781

➞ Concept of word embeddings used in feature representation.

[5] C. D. Manning, P. Raghavan, and H. Schütze, Introduction to Information Retrieval, Cambridge University Press, 2008.

🔗 https://nlp.stanford.edu/IR-book/

➞ Provides vector space model used in similarity matching.